

Attention is not all you need for robots

Final project

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Final Project: Autonomous and Gesture-Controlled Navigation Challenge

The objective of this project is to develop a comprehensive navigation and facial recognition system for a TurtleBot4 robot operating in a controlled environment. The project is divided into two distinct tasks. The robot must navigate through a circuit from a starting point (Point A) to a target destination (Point B) while avoiding obstacles, recognize a specific person from a photograph, and return to Point A to announce the identified person. The project is designed as a competition where teams will evaluate the effectiveness of their perception, decision-making, gesture-recognition, and navigation algorithms in the shortest possible time.

Objective

Develop a dual-mode robotic system capable of completing a two-way navigation and facial recognition circuit, operating both fully autonomously (Task A) and through video-based human gesture control (Task B).

Competition Environment

A circuit will be constructed featuring a starting zone (Point A) and a target zone (Point B). The environment will include:

- Obstacles of various sizes placed along the path.
- A photograph of a specific person located at Point B.

The final configuration of the circuit and the specific photograph used will only be revealed on the day of the competition. Students should design systems capable of generalizing to unseen environments and robust facial recognition, rather than memorizing a specific trajectory or a single image.

Tasks Description

The challenge consists of two mandatory tasks:

- **Task A (Autonomous Navigation):** The robot must autonomously navigate from Point A to Point B while avoiding obstacles. At Point B, it must use its camera to analyze the provided photograph. It must then autonomously navigate back to Point A and successfully output the name of the recognized person.
- **Task B (Gesture-Controlled Navigation):** The objective is identical to Task A (navigate A $\rightarrow$ B, recognize, return B $\rightarrow$ A, and announce). However, the robot cannot operate autonomously for navigation. Instead, it must be guided solely by recognizing human gestures and body movements via video. The use of traditional input devices (mouse, keyboard, joystick, etc.) is strictly prohibited.

A task is considered successfully completed only if the robot returns to Point A and correctly identifies the person in the photograph.

Attempts

Each team will have a maximum of three official attempts per task during the competition. Only the best attempt for each task will be considered for grading purposes.

Time Limit

Each attempt will have a maximum allowed execution time. The exact threshold will be announced after the final circuit has been validated. If the robot exceeds the allowed time, the attempt will be terminated automatically and no competition points will be awarded for that specific attempt.

Penalties

The objective of the competition is to demonstrate efficient navigation and accurate recognition. Time penalties will be applied for situations such as:

- Collisions with obstacles.
- Collisions with walls.
- Incorrect identification of the person in the photograph (which may result in a severe time penalty or a failed attempt, as defined in the lab).

The exact time penalty values (e.g., seconds added per collision) will be recorded and announced during the laboratory sessions prior to the competition. Any applicable penalties will be added to the final execution time used for scoring.

Human Intervention

The rules for human intervention depend strictly on the task being executed:

- **During Task A:** No human intervention is allowed once the attempt has started. Physically touching the robot, modifying the environment, or interacting with the execution system will result in immediate termination.
- **During Task B:** Human interaction is *required* but restricted strictly to visual cues (gestures, hand movements, body movements) captured by a camera. Physically touching the robot, modifying the environment, or using any manual controllers (mouse, keyboard, joystick, gamepads) will result in the immediate termination of the attempt.

Execution Platform

Teams may choose the software architecture that best fits their solution.

Examples include:

- Running all algorithms directly on the TurtleBot4.
- Running algorithms on an external computer connected through the network.
- Hybrid solutions combining onboard and external computation.

Students may use any software tools, frameworks, libraries, or pretrained models (such as facial recognition or pose estimation networks) that are relevant to the project.

Competition Scoring

The competition component contributes a maximum of 9 points to the final project grade. These 9 points are divided equally between the two tasks (4.5 points maximum for Task A, and 4.5 points maximum for Task B).

Task Evaluation Formula

For each task (A and B), teams that successfully complete the objective will compete for a score between 0 and 4.5 points based on their total execution time, including penalties.

For a successfully completed task, the score P_{task} is computed as:

$$P_{\text{task}} = 4.5 \cdot \frac{t_{\max} - t}{t_{\max} - t_{\min}}$$

where:

- P_{task} is the score obtained for that specific task.
- t is the team's completion time including time penalties.
- $t_{\min}$ is the best time recorded among all teams that completed the task.
- $t_{\max}$ is the worst time recorded among all teams that completed the task.

The total competition score P is the sum of the scores obtained in both tasks:

$$P = P_{\text{Task A}} + P_{\text{Task B}}$$

If only one team successfully completes a specific task, that team automatically receives the maximum 4.5 points available for that task. If a team fails to complete a task, their score for that specific task will be 0.

Deliverables

Presentation

Each team must present both the technical aspects of their proposed solution and the results obtained during the competition.

The presentation should focus on:

- Overall system architecture.
- Facial recognition implementation.
- Gesture recognition and teleoperation strategy (Task B).
- Autonomous navigation strategy (Task A).
- Sensor integration.
- Models and algorithms employed.
- Design decisions and engineering trade-offs.
- Experimental results.
- Lessons learned during development.

Students are expected not only to demonstrate that their system works, but also to explain and justify the algorithms and methodologies used throughout the project. The presentation should be organized as a technical discussion of the proposed solution rather than a simple demonstration of the final system. Time is strictly limited to 8 to 10 minutes per group.

Submission

Each team must upload both the project report and the presentation slides to Canvas before the deadline announced in class. Failure to submit the required materials will result in the inability to present the project. The report should focus primarily on the methodology used by the team. In particular, students are expected to:

- Describe the overall architecture of the proposed solution.
- Explain the computer vision algorithms employed for both facial and gesture recognition.
- Explain the autonomous navigation and decision-making strategies.
- Justify the selection of models, algorithms, and design choices.
- Discuss limitations and possible future improvements.

The report should emphasize technical explanations and system design rather than source code. If students wish to include implementation details or code snippets, these should be placed in an appendix at the end of the document.

Laboratory Organization

The final project will be evaluated using the available TurtleBot4 robots. Because the number of robots is limited, the number of project groups that can be evaluated simultaneously is also limited. In general, no more than five groups will be assigned to the same laboratory section. The final number of groups per section may

vary depending on robot availability and laboratory logistics. Additional details regarding group distribution, robot availability, and laboratory scheduling will be coordinated during the semester.

Rubric

The final project is graded out of 20 points, heavily emphasizing both the technical deliverables (11 points) and the competition performance (9 points).

- Report and technical documentation: 4 pts
- Presentation: 4 pts
- Questions: 3 pts
- Competition score: 9 pts

Group Work

Group assignments may be adjusted if required by laboratory capacity constraints or robot availability.

Final consideration

All groups must create a video containing your competition; it means, when your robot competes in the challenge for both Task A and Task B. This video has to be uploaded to the classroom shared drive folder.

Finally, this challenge is part of the grade; therefore, students must be kind and also show respect during the competition. Good luck folks!!!